Inference for linear regression models

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

> What is our population model?

Activity

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

> How would I interpret the estimated slope?

Activity

Your parents, worried about the nutritional value of your breakfast, believe that there will be a positive relationship between the amount of sugar in a serving, and the number of calories per serving.

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

> What are our null and alternative hypotheses?

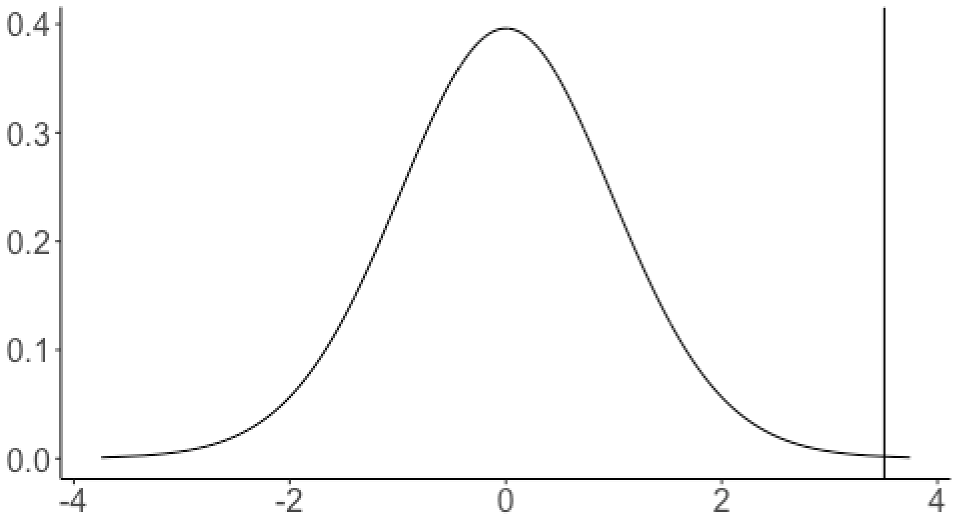
Activity

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 > 0$$

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

> What is our test statistic? What are the degrees of freedom?

Activity



> What area under the curve corresponds to my p-value?

A new question

$$\text{Calories} = \beta_0 + \beta_1 \text{Sugar} + \varepsilon$$

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

Here $\widehat{\beta}_1 = 2.48$. Will $\widehat{\beta}_1$ be exactly equal to the true value β_1?

A new question

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```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
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```

> Here $\widehat{\beta}_1 = 2.48$. Will $\widehat{\beta}_1$ be exactly equal to the true value β_1?

No -- our estimate comes from one sample of data, and there is variability from sample to sample.

> How can I account for this variability when reporting an estimate of β_1?

Interval estimates

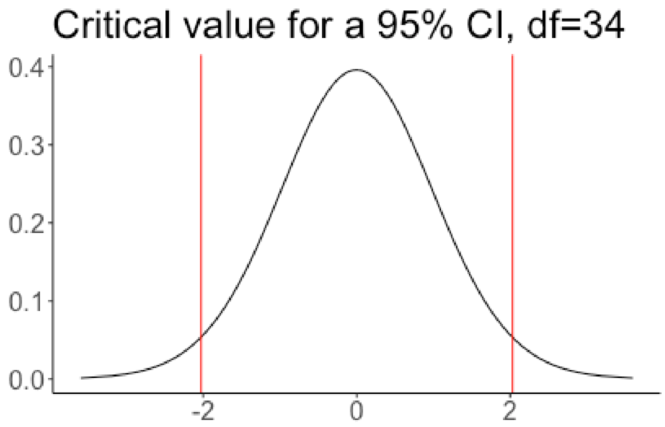
Point estimate: a single "best guess" value of a parameter. The estimated slope $\widehat{\beta}_1$ is a point estimate of β_1

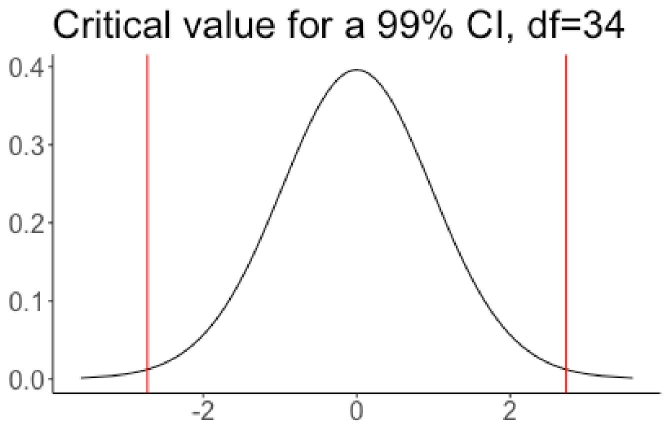
Interval estimate: a "reasonable range" of possible values for a parameter. A *confidence interval* is an interval estimate

> How would I construct a confidence interval for β_1?

The critical value t^*

To construct a $(1-\alpha)\cdot 100\%$ confidence interval, t^* is the value such that the area in the right tail of the t_{n-2} distribution is $\alpha/2$





Computing t^*

Example: for a 95% confidence interval and 34 degrees of freedom, t^* = 2.03

```r
qt(0.025, df=34, lower.tail=FALSE)
```

```
## [1] 2.032245
```

Example: for a 99% confidence interval and 34 degrees of freedom, t^* = 2.73

```r
qt(0.005, df=34, lower.tail=FALSE)
```

```
## [1] 2.728394
```

Confidence interval

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

95% confidence interval for β_1:

Confidence interval

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

95% confidence interval for β_1:

$$2.48 \pm 2.03(0.71) = (1.04, 3.92)$$

> How do I interpret this confidence interval?

Confidence interval

```
Coefficients:
            Estimate Std. Error
(Intercept)  87.4277     5.1627
Sugar         2.4808     0.7074
```

95% confidence interval for β_1:

$$2.48 \pm 2.03(0.71) = (1.04, 3.92)$$

Interpretation: We are 95% confident that a one-gram increase in sugar per serving of breakfast cereal is associated with an increase of between 1.04 and 3.92 calories per serving.

> What does it mean to be "95% confident"?

Activity

Work on the activity (handout), then we will discuss as a class.

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(..., df = ..., lower.tail=FALSE)
```

> What do I fill in to calculate the critical value t^*?

Activity

Data on 116 sparrows. Want to make a 98% confidence interval.

```
qt(0.01, df = 114, lower.tail=FALSE)
```

```
## [1] 2.359504
```

Activity

```
              Estimate Std. Error
(Intercept)   8.75465    1.39601
Weight        1.31341    0.09756
```

$t^* = 2.36$

> What is my 98% confidence interval?

Activity

```
              Estimate Std. Error
(Intercept)  8.75465    1.39601
Weight       1.31341    0.09756
```

$t^* = 2.36$

$$1.31 \pm 2.36(0.098) = (1.08, 1.54)$$

> Your friend claims that there is a 2% chance that the true slope β_1 is greater than 1.54 or less than 1.08. Is your friend correct?